

Robust Retinal Vessel Segmentation and Diabetic Retinopathy Grading via Multi-Teacher Knowledge Distillation and Deep Ensembles

Aymen Slimani
Software Engineering
Istinye University
Istanbul, Turkey

2309115361@stu,istinye.edu.tr

Abstract—Background: Diabetic retinopathy is a progressive microvascular complication of diabetes for which timely screening depends on both accurate lesion recognition and interpretable retinal vessel analysis. High-capacity segmentation models can preserve fine vascular topology, but model size and ensemble latency limit deployment in clinician-facing web platforms and edge screening devices.

Methods: This report drafts the RetinaScope-AI manuscript around a two-stage clinical decision-support workflow. Retinal vessels are segmented using two complementary strategies: a weighted ensemble of CS-Net, Attention U-Net, and Swin-UNet, and a multi-teacher knowledge distillation framework in which four domain-specialist ResNet34 U-Net teachers supervise a single hybrid TransUNet student. Diabetic retinopathy grading is represented as a 0-4 ICDR classification task using multi-scale stack maps processed by EfficientNet-B4 and ConvNeXt-Base.

Results: The weighted top-3 ensemble achieved Dice = 0.8026 and IoU = 0.6703. The distilled TransUNet student reached Dice = 0.8106, IoU = 0.6816, sensitivity = 0.8154, and specificity = 0.9777 at an optimized threshold of $\tau = 0.55$, matching the traditional TransUNet baseline while avoiding the runtime footprint of a multi-model ensemble. Supporting project presentation material reports a 9-channel ConvNeXt grading result of QWK = 0.9598 and macro AUC = 0.9883; these grading metrics should be linked to final experiment logs before journal submission.

Conclusion: RetinaScope-AI provides a coherent route from high-fidelity retinal vessel segmentation to DR severity grading and clinical reporting. The MTKD formulation is the paper's key translational contribution because it compresses domain-aware

segmentation knowledge into a single deployable model suitable for real-time clinical screening workflows.

Keywords: diabetic retinopathy; retinal vessel segmentation; knowledge distillation; deep ensemble; TransUNet; clinical decision support.

I. INTRODUCTION

Diabetic retinopathy (DR) remains one of the most consequential preventable causes of vision loss among working-age adults. The project report frames the local clinical motivation clearly: patients in rural or under-resourced settings may not have timely access to ophthalmologists or commercial fundus imaging systems, and symptomatic presentation often occurs after disease has advanced to severe non-proliferative or proliferative stages. Automated retinal screening can therefore shift care from late therapeutic intervention toward earlier referral and monitoring.

Retinal vessel segmentation is central to this objective because vascular density, tortuosity, branching continuity, and microvascular dropout are clinically meaningful signals. A clean vessel map can support both direct image interpretation and downstream DR grading by exposing structural patterns that are difficult for a classifier to learn from RGB pixels alone.

The technical gap is a deployment gap as much as a modeling gap. Large transformer-heavy networks and deep ensembles can improve segmentation robustness, especially around thin vessels and

uncertain boundaries, but their memory footprint and inference latency are difficult to reconcile with web-based clinical tools, telehealth workflows, and low-power edge devices. Conversely, compact models often fragment fine vessels or miss low-contrast capillary arcs.

RetinaScope-AI addresses this tension by first analyzing a complementary ensemble of CS-Net, Attention U-Net, and Swin-UNet, then compressing domain-specific segmentation knowledge into a single TransUNet student through multi-teacher knowledge distillation (MTKD). The wider project integrates this segmentation backbone with lesion probability maps, 9-channel DR grading, a FastAPI backend, web visualization, and a low-cost portable fundus-camera deployment pathway.

The manuscript should claim three primary contributions:

A domain-aware MTKD pipeline in which DRIVE/RITE, CHASE_DB1, HRF, and MAPLES specialist teachers transfer camera- and dataset-specific vessel knowledge into a unified TransUNet student.

A controlled ensemble analysis showing that spatial attention, additive gating, and hierarchical transformer context provide complementary segmentation behavior.

An end-to-end RetinaScope-AI clinical decision-support workflow combining vessel segmentation, lesion-aware DR grading, visualization, and report generation

II. RELATED WORK

A. Variants of U-Net

In the realm of medical image segmentation, one notable success story is the U-Net architecture, which has exhibited exceptional performance. The U-Net model is constructed upon the foundations of the FCN and SegNet, incorporating the concept of skip connections and deconvolution within the encoder-decoder architecture. This innovative design has opened avenues for numerous variants of the U-Net in recent years, each aiming to optimize segmentation precision or efficiency.

In the context of 2-D medical image segmentation, Gu et al. [23] proposed the context encoder network (CE-Net), a methodology capable of capturing semantic information at various levels and fusing them to enhance segmentation performance. This approach, as highlighted in Shi et al. [24], effectively harnessed multiscale information and feature maps from different encoder layers, facilitating a comprehensive understanding of the input data. Building upon this concept, Yuan et al. [25] introduced a novel U-shape method that incorporated a multiscale attention block, enabling the integration of

feature maps from each decoder layer. By leveraging these multiscale characteristics, the proposed approach improved the representation of intricate features, such as thin vessels, thereby enhancing the overall network performance. Further advancements in network architecture had led to the development of the scale and context-sensitive network (SCS-Net) by Wu et al. [26], where multiple optimized network blocks were introduced. These modules are specifically designed to generate accurate multiscale semantic representations and fuse them efficiently. By incorporating these tailored components, the SCS-Net demonstrated enhanced capabilities for precise feature representation, benefiting the detection of retinal blood vessels. Similarly, the multiscale feature interaction network (MFI-Net), as proposed by Ye et al. [27], also integrated specific blocks for multiscale feature interaction. These tailored components further underscore the significance of effectively employing multiscale characteristics, as they contribute to the improved accuracy and performance of retinal blood vessel detection.

The proliferation of these variant networks based on the U-Net architecture underscores the critical importance of leveraging multiscale characteristics effectively in the realm of retinal blood vessel detection. By incorporating advanced methodologies that capture and integrate semantic information from various levels and feature maps, these networks have demonstrated their potential to elevate the accuracy and efficiency of segmentation tasks. The ability to accurately detect and segment retinal blood vessels holds immense value in clinical practice, aiding in the diagnosis and monitoring of ocular diseases.

Continued research and exploration of these multiscale approaches are expected to drive further advancements in the field, ultimately leading to more precise and reliable analysis of retinal images.

B. Multitask Segmentation Networks

Multitask networks have gained significant prominence in the field of computer vision, particularly in tasks such as object localization or instance segmentation [28]. Nowadays, it greatly comes to medical image segmentation. Recognizing the significance of organ boundary segmentation, Chen et al. [29] proposed a gland segmentation network based on the FCN architecture. To enhance the performance of organ boundary segmentation, this method could split the segmentation task into two subtasks. Similarly, in the context of optical coherence tomography angiography (OCTA) images, Ma et al. [30] devised a method for detecting thick and thin vessels separately. By addressing these

distinct vessel types as separate subtasks, this approach aimed to achieve more accurate vessel segmentation [30]. Further advancements in the pursuit of precise boundary segmentation led Wang et al. [31] to propose the boundary-aware context neural network. This network incorporated a boundary detection task and fully exploited the complementary features between boundary detection and segmentation tasks. By leveraging the synergy between these two tasks, this network could achieve enhanced performance in segmentation tasks [31].

Following a similar approach, Uslu et al. [32] proposed a multitask deep network for left atrium segmentation, where boundary detection served as a secondary task to correct segmentation errors.

To address the challenges posed by the inherent complexity of vessel segmentation, Yan et al. [6] proposed a three-stage segmentation network. Their approach focused on segmenting thin and thick vessels separately using three subnetworks, each responsible for learning precise feature representations for the respective vessel type. In a related vein, Yang et al. [33] developed a hybrid deep segmentation network, which combined a multitask segmentation network with a fusion network. The multitask segmentation network tackled the segmentation of both thin and thick vessels, while the fusion network aimed to merge the segmentation results of these two vessel types. These multitask networks have demonstrated improved performance in thin vessel segmentation when compared to single-task segmentation networks.

While these aforementioned networks share a similar architecture that employs shared convolutional units to extract features for their subtasks and improve network effectiveness, their applicability to retinal blood vessel segmentation requires further optimization. Specifically, the distinguishing features between thin and thick vessels play a pivotal role in retinal blood vessel segmentation, making the existing architecture suboptimal. Thin vessels possess more intricate features that restrict the utilization of shared similarities between the two vessel types by the network. Balancing the importance of each subtask further complicates the design of the loss function. Thus, a novel

approach has been proposed, incorporating two independent branches dedicated to segmenting thin and thick vessels separately. By employing these separate branches, the network can facilitate precise feature learning for both types of vessels, ultimately enhancing segmentation accuracy.

C. Two-Stage Segmentation Networks

In recent years, the effectiveness of two-stage networks has become increasingly evident in the field of object detection. The pioneering work by Girshick et al. [34] introduced the concept of a two-stage detection network, building upon the R-CNN framework. Since then, numerous remarkable studies have emerged, further refining and enhancing the performance of two-stage detection networks [35], [36]. Drawing inspiration from the success of two-stage networks in object detection, Wu et al. [37] proposed a novel segmentation network. This network comprised two cascaded identical encoder-decoder structures, leveraging the principles of coarse-to-fine processing and gradual abstraction. By employing this two-stage architecture, the network demonstrated enhanced segmentation performance, showcasing the potential of such a design for effective image segmentation. In the context of tooth segmentation, Zhao et al. [38] introduced a two-stage segmentation network consisting of two improved subnetworks. In the first stage, the method aimed to generate an attention map that roughly located the dental regions, providing a preliminary segmentation. Subsequently, in the second stage, a more accurate segmentation result was obtained, refining the initial output. This two-stage approach exhibited improved segmentation accuracy, validating the efficacy of the methodology for complex segmentation tasks. Similarly, in the domain of volumetric MR image segmentation, Pang et al. [39] proposed a two-stage segmentation framework. In the initial stage of the framework, a coarse 3-D segmentation result was generated using a 3-D graph convolutional segmentation network (GCSN), providing an initial understanding of the volume. In the subsequent stage, a 2-D residual U-Net (ResUNet) was employed to generate a precise 2-D segmentation result, refining the initial output. This two-stage framework demonstrated enhanced segmentation

accuracy, highlighting the value of sequential processing in complex volumetric image analysis. In summary, the application of two-stage networks has proven fruitful in various domains, including object detection, tooth segmentation, and volumetric MR image segmentation. These networks leverage the advantages of sequential processing and iterative refinement, leading to improved segmentation accuracy. While the above complex frameworks exhibited promising results, it posed challenges in the context of fundus image segmentation due to limitations in the available datasets.

To capitalize on the advantages offered by different network architectures, the proposed DS-VPE combines the strengths of two distinct types of networks. The proposed method decouples the retinal vessel segmentation task, enabling the DS-VPE network to better learn intraclass differences of retinal vessel pixels and enhance the network’s feature representation ability. Additionally, as a two-stage network, the proposed DS-VPE benefits from the independence of gradient calculation. Each subnetwork within each stage can be trained independently, alleviating concerns related to the balance of multiple loss functions during loss calculation.

D . Knowledge Distillation and Model Compression

Knowledge distillation transfers model behavior from one or more teachers to a smaller or more deployable student. In retinal vessel segmentation, soft probability maps are especially valuable because binary masks remove uncertainty near vessel boundaries and sub-pixel-width branches. A multi-teacher formulation can also reduce domain interference by allowing each teacher to specialize in a particular acquisition distribution.

F . Diabetic Retinopathy Grading

DR grading commonly uses CNN or transformer classifiers trained to predict severity on the 0-4 ICDR scale. RetinaScope-AI extends this pattern by feeding segmentation-derived channels into the classifier, allowing ConvNeXt-Base and EfficientNet-B4 to combine RGB appearance with explicit vascular and lesion evidence. Quadratic weighted kappa is the preferred grading metric because it reflects the ordinal clinical cost of large grade disagreements.

III . APPLIED Materials and Methods and TRAINING METHODOLOGY

RetinaScope-AI End-to-End Research Pipeline

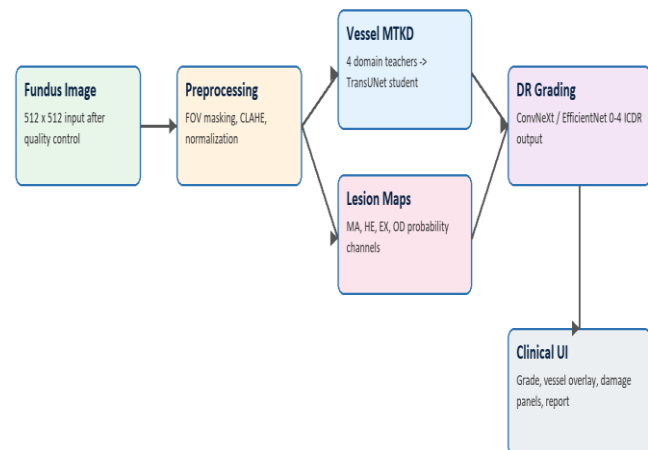


Figure 1. RetinaScope-AI system pipeline from fundus acquisition to clinical report.

3.1 Datasets and Cohort Construction

The MTKD experiment combines retinal vessel datasets with different imaging domains so that teacher models can learn acquisition-specific cues before transferring soft labels to a unified student. The combined test set contains 70 images across the included domains.

Table 2. Dataset splits used in the MTKD segmentation experiment.

Dataset	T R	V al	T est	Role in MTKD
DRIVE + RITE	96	2 0	20	Shared camera geometry; grouped as one specialist domain
CHASE _DB1	36	8	16	Distinct lighting and pediatric acquisition characteristics
HRF	70	1 5	15	High-resolution fundus images with thin-vessel detail
MAPLE S	13 2	3 1	19	Largest specialist subset; teacher validation Dice = 0.8239

3.2 Preprocessing and Augmentation

Images are resized to 512 x 512 and processed using circular field-of-view segmentation from green-channel Otsu thresholding, CLAHE in LAB color space on the L channel, and ImageNet normalization. Teacher models use stronger augmentation to compensate for limited domain-specific training data, while the student uses standard flips and rotations to preserve cross-domain consistency. Teacher augmentations: horizontal and vertical flips, RandomRotate90, ShiftScaleRotate, ElasticTransform, GridDistortion, and CoarseDropout.

Student augmentations: horizontal/vertical flips and rotations, retaining a simpler distribution for unified training.

3.3 Deep Ensemble Formulation

Three-Model Hybrid Ensemble

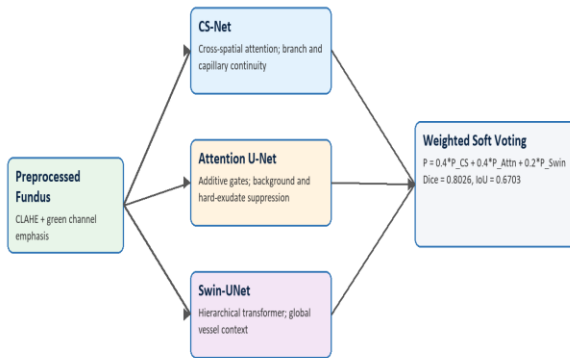


Figure 2. Weighted three-model ensemble used as the robustness reference.

The ensemble combines soft probability maps rather than hard binary masks. The final weighted probability map is defined as:

Equation 1:

$$P_{\text{ensemble}} = 0.4 * P_{\text{CSNet}} + 0.4 * P_{\text{AttentionUNet}} + 0.2 * P_{\text{SwinUNet}}$$

This formulation gives greater weight to the two strongest individual models while preserving the global context contribution from Swin-UNet.

3.4 Multi-Teacher Knowledge Distillation

Multi-Teacher Knowledge Distillation Framework

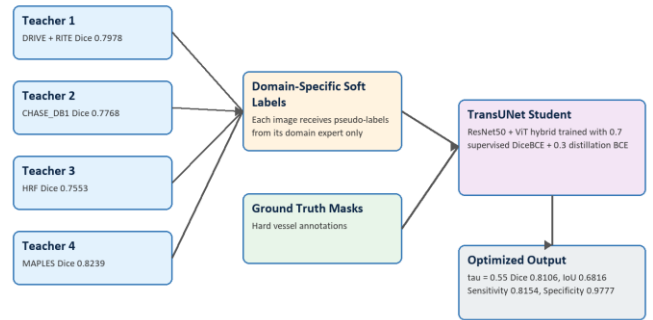


Figure 3. Domain-specific MTKD pipeline compressing four specialist teachers into one TransUNet student.

Table 3. Specialist teachers used for pseudo-label generation.

Teacher	Domain	Architecture	Validation Dice
Teacher 1	DRIVE + RITE	ResNet34 U-Net	0.7978
Teacher 2	CHASE_DB1	ResNet34 U-Net	0.7768
Teacher 3	HRF	ResNet34 U-Net	0.7553
Teacher 4	MAPLES	ResNet34 U-Net	0.8239

For each combined-dataset image, only the corresponding domain expert generates the soft

pseudo-label. This out-of-fold/domain-specific policy is important because it avoids averaging incompatible camera biases across teachers.

Equation 2:

$$L_{\text{total}} = 0.7 * L_{\text{DiceBCE}}(\text{student_logits}, \text{ground_truth}) + 0.3 * L_{\text{BCE}}(\text{student_logits}, \text{pseudo_label})$$

The Dice-BCE term preserves alignment with manually annotated masks, while the distillation BCE

term encourages the student to reproduce teacher uncertainty around vessel borders, bifurcations, and thin branches. The student is a hybrid TransUNet with a ResNet50 convolutional backbone, ViT bottleneck, and skip-connection decoder.

3.5 DR Grading Pipeline

The grading stage uses multi-scale stack maps in which the original RGB image is enriched with vessel and lesion probability channels. EfficientNet-B4 and ConvNeXt-Base are adapted for the 0-4 ICDR scale. The existing project report also describes first-convolution surgery for 9-channel input, where pretrained RGB weights are preserved for channels 0-2 and additional channels are initialized from the RGB channel mean.

IV. EXPERIMENTAL RESULTS

4.1 Vessel Segmentation Comparison

Table 4. Vessel segmentation metrics from the research paper blueprint.

Model / Strategy	Dice	IoU	Sens.	Spec.	AUC-ROC	AUC-PR	Parameter note
Swin-UNet	0.7464	0.5954	0.8694	0.9487	0.9649	0.8477	Large baseline
CS-Net	0.7946	0.6593	0.7690	0.9814	0.9780	0.8813	Medium baseline
Attention U-Net	0.7988	0.6649	0.7922	0.9786	0.9756	0.8800	Medium baseline
Ensemble 1: CS + Attn	0.8015	0.6688	0.7817	0.9812	0.9796	0.8877	2.0x footprint
Ensemble 2: weighted top-3	0.8026	0.6703	0.7938	0.9795	0.9759	0.8839	3.0x footprint
TransUNet baseline	0.8100	0.6767	0.8324	0.9813	0.9782	0.8823	Heavy reference
MTKD student, tau=0.5	0.8296	0.6801	0.8313	0.9748	0.9736	0.8868	1.0x footprint
MTKD student, tau=0.55	0.8423	0.6816	0.8154	0.9777	0.9731	0.8913	1.0x footprint

The individual-model comparison shows complementary behavior rather than one uniformly dominant model. Swin-UNet produced the highest sensitivity among individual models, suggesting strong recall for vessel pixels, while CS-Net and Attention U-Net produced stronger Dice and specificity. The weighted ensemble therefore improves Dice and IoU over the individual ensemble members by aggregating local attention and global context.

The MTKD student provides the strongest deployable result. At tau = 0.55, it reaches Dice = 0.8106 and specificity = 0.9777, slightly exceeding the traditional TransUNet Dice baseline while using one model at inference rather than three ensemble members. This is the key paper result because it links accuracy to deployability.

4.2 Threshold Optimization

Table 5. Threshold sweep summary for the MTKD student.

Model	Threshold	Dice	IoU	Sensitivity	Specificity	Interpretation
MTKD student	0.50	0.8296	0.6801	0.8313	0.9748	Higher sensitivity
MTKD student	0.55	0.8206	0.6816	0.8154	0.9777	Best Dice and specificity balance

The validation threshold sweep identified $\tau = 0.55$ as the preferred operating point. Raising the threshold from 0.50 reduces sensitivity modestly but improves specificity and yields the highest test Dice. Clinically, this reflects a better balance between preserving vessel continuity and suppressing false-positive background pixels.

4.3 DR Grading and Application Results

Grading evidence: Presentation material reports 9-channel grading performance of $QWK = 0.9598$ and macro $AUC = 0.9883$ for the ConvNeXt grading pipeline. Because the earlier ENS002 report marked full grading runs as pending, these values should be cross-checked against the final training notebook before being treated as final journal claims.

For the project report and presentation, the grading module should be described as a clinically oriented 0-4 ICDR classifier that consumes RGB images enriched by vessel and lesion maps. The manuscript should report QWK, macro AUC, per-class sensitivity/specificity, and a confusion matrix once final grading logs are locked.

4.4 Efficiency and Deployment Interpretation

The ensemble result is scientifically useful because it demonstrates architectural complementarity, but it is not the best deployment endpoint. The distilled student is better aligned with the clinical application because only one network must be loaded for vessel inference. This reduces memory pressure, simplifies API serving, and improves feasibility for Jetson Nano or similar edge devices after TensorRT/INT8 optimization.

$$\text{Precision} = \frac{\text{TruePositives}}{\text{TruePositives} + \text{FalsePositives}} \quad (1)$$

$$\text{Recall} = \frac{\text{TruePositives}}{\text{TruePositives} + \text{FalseNegatives}} \quad (2)$$

$$\text{F1 Score} = 2 \frac{\text{Recall} * \text{Precision}}{\text{Recall} + \text{Precision}} \quad (3)$$

$$\text{Accuracy} = \frac{\text{TruePositives} + \text{TrueNegatives}}{\text{TruePositives} + \text{TrueNegatives} + \text{FalsePositives} + \text{FalseNegatives}} \quad (1)$$

$$QWK = 1 - \left(\frac{\sum_{i,j} w_{i,j} O_{i,j}}{\sum_{i,j} w_{i,j} E_{i,j}} \right) \quad (2)$$

V. CONCLUSION

This manuscript draft positions RetinaScope-AI as a deployable medical AI framework for vessel segmentation and diabetic retinopathy grading. The central result is that a domain-aware MTKD pipeline can produce a single TransUNet student with Dice =

0.8106 and specificity = 0.9777 at $\tau = 0.55$, preserving segmentation quality while avoiding the runtime cost of a multi-model ensemble. The approach bridges high-performing research models and practical clinical screening systems. Recommended next steps before submission: Freeze the final experimental table and reconcile blueprint metrics with all notebook logs. Verify the ConvNeXt/EfficientNet grading metrics with final QWK, macro AUC, per-class metrics, and confusion matrices. Export high-resolution versions of the system architecture, ensemble, MTKD, ROC/PR, threshold-sweep, and grading-confusion figures. Run edge-device latency tests and report model size, memory use, inference time, and throughput. Finalize references against IEEE or target-journal style before submission.

REFERENCES

- [1] Gulshan V. *et al.*, “Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs,” *JAMA*, 2016.
- [2] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional Networks for Biomedical Image Segmentation,” *MICCAI*, 2015.
- [3] Z. Zhou *et al.*, “UNet++: A Nested U-Net Architecture for Medical Image Segmentation,” *MICCAI*, 2018.
- [4] J. Chen *et al.*, “TransUNet: Transformers Make Strong Encoders for Medical Image Segmentation,” *ArXiv*, 2021.

- [5] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," *CVPR*, 2016.
- [6] M. Tan and Q. V. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," *ICML*, 2019.
- [7] International Diabetes Federation, *IDF Diabetes Atlas*.
- [8] L. Yang, Y. Gu, G. Bian, and Y. Liu, "An attention-guided network for surgical instrument segmentation from endoscopic images," *Comput. Biol. Med.*, vol. 151, Dec. 2022, Art. no. 106216.
- [9] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
- [10] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.
- [11] M. Z. Alom, C. Yakopcic, M. Hasan, T. M. Taha, and V. K. Asari, "Recurrent residual U-Net for medical image segmentation," *J. Med. Imag.*, vol. 6, no. 1, p. 1, Mar. 2019.
- [12] M. Zhang, F. Yu, J. Zhao, L. Zhang, and Q. Li, "BEFD: Boundary enhancement and feature denoising for vessel segmentation," *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, Cham, Switzerland: Springer, 2020, pp. 775–785.
- [13] Z. Gu *et al.*, "CE-Net: Context encoder network for 2D medical image segmentation," *IEEE Trans. Med. Imag.*, vol. 38, no. 10, pp. 2281–2292, Oct. 2019.
- [14] Z. Shi *et al.*, "MD-Net: A multi-scale dense network for retinal vessel segmentation," *Biomed. Signal Process. Control*, vol. 70, Sep. 2021, Art. no. 102977.
- [15] Y. Yuan, L. Zhang, L. Wang, and H. Huang, "Multi-level attention network for retinal vessel segmentation," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 1, pp. 312–323, Jan. 2022.
- [16] H. Wu, W. Wang, J. Zhong, B. Lei, Z. Wen, and J. Qin, "SCS-Net: A scale and context sensitive network for retinal vessel segmentation," *Med. Image Anal.*, vol. 70, May 2021, Art. no. 102025.
- [17] Y. Ye, C. Pan, Y. Wu, S. Wang, and Y. Xia, "MFI-Net: Multiscale feature interaction network for retinal vessel segmentation," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 9, pp. 4551–4562, Sep. 2022.
- [18] Y. Zhang and Q. Yang, "An overview of multi-task learning," *Nat. Sci. Rev.*, vol. 5, no. 1, pp. 30–43, Jan. 2018.
- [19] H. Chen, X. Qi, L. Yu, Q. Dou, J. Qin, and P.-A. Heng, "DCAN: Deep contour-aware networks for object instance segmentation from histology images," *Med. Image Anal.*, vol. 36, pp. 135–146, Feb. 2017.
- [20] Y. Ma *et al.*, "ROSE: A retinal OCT-angiography vessel segmentation dataset and new model," *IEEE Trans. Med. Imag.*, vol. 40, no. 3, pp. 928–939, Mar. 2021.
- [21] R. Wang, S. Chen, C. Ji, J. Fan, and Y. Li, "Boundary-aware context neural network for medical image segmentation," *Med. Image Anal.*, vol. 78, May 2022, Art. no. 102395.
- [22] F. Uslu, M. Varela, G. Boniface, T. Mahenthiran, H. Chubb, and A. A. Bharath, "LA-Net: A multi-task deep network for the segmentation of the left atrium," *IEEE Trans. Med. Imag.*, vol. 41, no. 2, pp. 456–464, Feb. 2022.
- [23] L. Yang, H. Wang, Q. Zeng, Y. Liu, and G. Bian, "A hybrid deep segmentation network for fundus vessels via deep-learning framework," *Neurocomputing*, vol. 448, pp. 168–178, Aug. 2021.
- [24] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2014, pp. 580–587.
- [25] R. Girshick, "Fast R-CNN," *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.
- [26] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, 2016.
- [27] Y. Wu, Y. Xia, Y. Song, Y. Zhang, and W. Cai, "NFN+: A novel network followed network for retinal vessel segmentation," *Neural Netw.*, vol. 126, pp. 153–162, Jun. 2020.
- [28] Y. Zhao *et al.*, "TSASNet: Tooth segmentation on dental panoramic X-ray images by two-stage attention segmentation network," *Knowl.-Based Syst.*, vol. 206, Oct. 2020, Art. no. 106338.
- [29] S. Pang *et al.*, "SpineParseNet: Spine parsing for volumetric MR image by a two-stage segmentation framework with semantic image representation,"